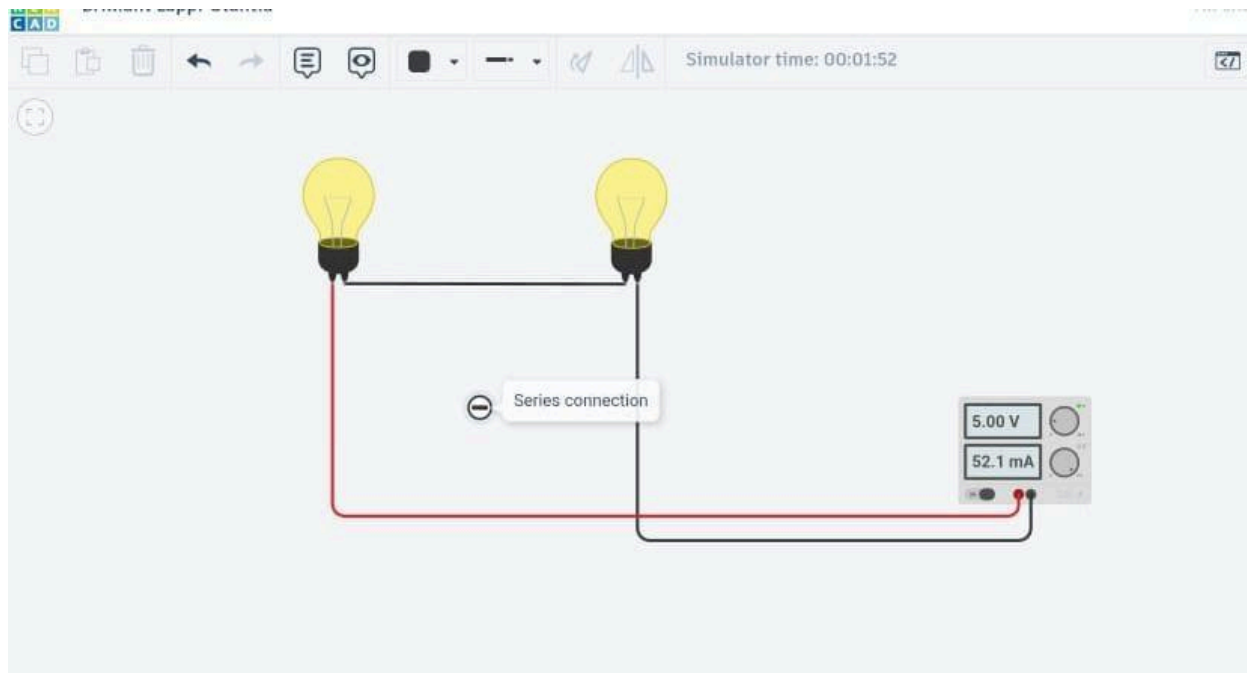
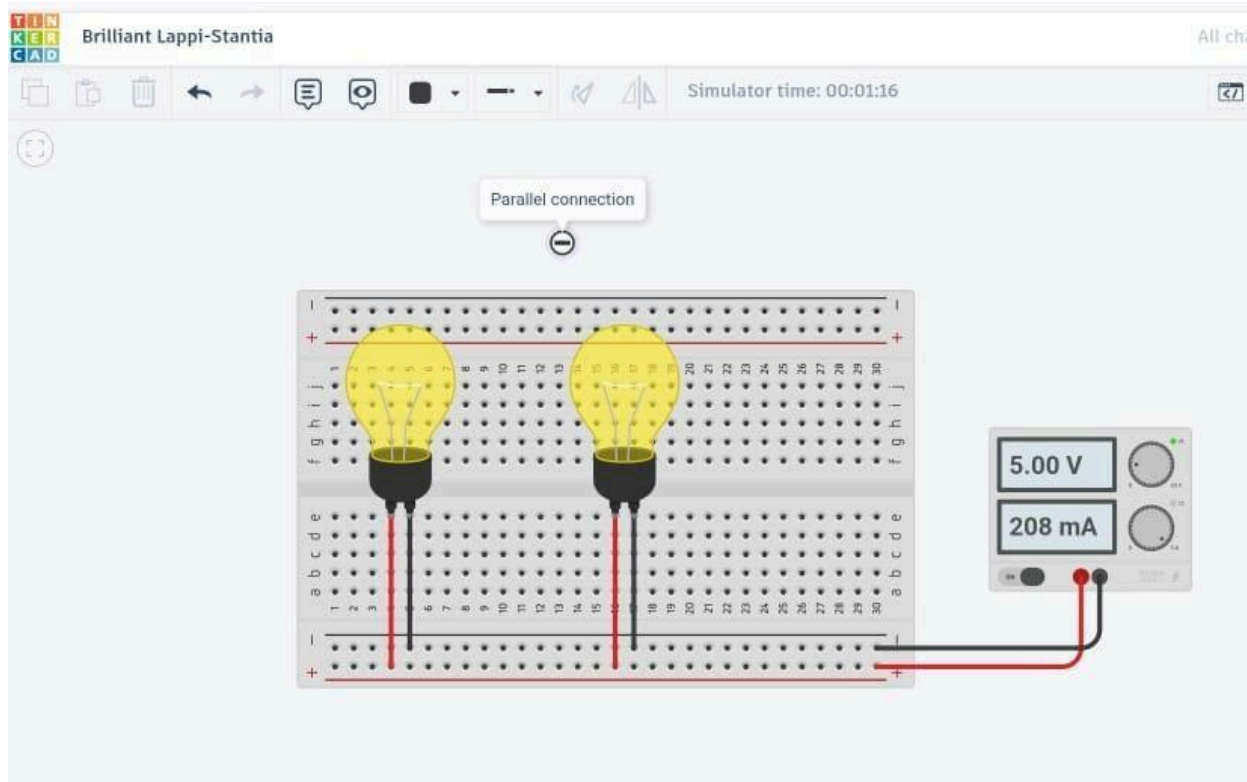


Series and parallel connection.

Series connection



Parallel connection.



Feature	Series Connection	Parallel Connection
Current (I)	Same current flows through all components	Current splits between branches
Voltage (V)	Voltage is divided among components	Same voltage across all components
Resistance (R)	Total resistance increases ($R_1 + R_2 + \dots$)	Total resistance decreases

Technical Calculation Document: Motor Analysis

Motor Model: TFK-280SA-22125

Objective: To determine the operational parameters for normal use versus the calculated limits for intentional "override" (failure).

1. Fundamental Motor Constant

The internal resistance R is the physical opposition the motor windings offer to the flow of current. This is calculated based on the "Stall" (stationary) state where the motor draws maximum current.

- **Internal Resistance :**

$$R = \frac{V_{nominal}}{I_{stall}} = \frac{6.0V}{2.6A} \approx 2.31 \Omega$$

2. Normal Operating Calculations

These calculations represent the motor functioning within the manufacturer's safety specifications.

- **No-Load Power (P_{idle}):** The power consumed when the motor is spinning freely.

$$P = 6.0V \times 0.09A = 0.54 \text{ W}$$

- **Maximum Rated Load Power (P_{max}):** The upper limit of safe power consumption before overheating begins.

$$P = 6.0V \times 3.2A \text{ (Max Stall)} = 19.2 \text{ W}$$

- **Rated Speed:** $7500 \pm 10\%$ rpm.

3. Override (Failure) Calculations

These calculations predict the "burn-out" scenarios where electrical and thermal limits are exceeded.

A. 12V Scenario (Critical Limit)

Doubling the voltage significantly increases the heat generated within the motor.

- **Override Current:**

$$I = \frac{12.0\text{V}}{2.31\ \Omega} \approx 5.19\ \text{A}$$

- **Total Heat Dissipation:**

$$P = 12.0\text{V} \times 5.19\text{A} \approx 62.28\ \text{W}$$

-

Result: This power level is over 3 times the maximum rated load. The insulation on the copper windings will begin to melt within seconds.

B. 18V Scenario (Total Destruction)

- **Critical Current:**

$$I = \frac{18.0\text{V}}{2.31\ \Omega} \approx 7.79\ \text{A}$$

- **Critical Power:**

$$P = 18.0\text{V} \times 7.79\text{A} \approx 140.22\ \text{W}$$

-

Result: The windings will glow red-hot almost instantly, likely causing an internal short circuit and potential fire hazard for plastic components.

Summary Table

Mode	Voltage	Current (Stall)	Power	Status
Normal	6.0V	2.6A	15.6 W	Safe
Override	12.0V	5.2A	62.3 W	High Risk
Critical	18.0V	7.8A	140.2 W	Instant Failure

Note: Intentional failure of a motor produces toxic fumes from melting insulation. Ensure proper ventilation if conducting these experiments.

